Blanching Unit



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Blanching Unit; Vegetable bleacher

What it is used for

A blanching unit briefly heats vegetables or other suitable food products in hot water or steam before cooling or further processing. Correct temperature, time, loading, and hygiene control help maintain product quality, support enzyme inactivation, and reduce food safety risks.

Main parts

- Heating chamber or tank holds the blanching medium or food product.
- Heating element, burner, or steam inlet supplies heat.
- Product basket, conveyor, or loading tray holds the food.
- Temperature sensor or thermometer monitors process temperature.
- Timer or controller sets the blanching cycle where fitted.
- Water inlet and drain support filling and emptying.
- Lid or cover helps retain heat and reduce contamination.
- Cooling or transfer arrangement moves the product to the next process where included.

Before you start

1. Only trained food-processing personnel should operate the blanching unit.
2. Read the operating manual and confirm the product-specific temperature, time, and loading requirements.
3. Inspect the tank, basket, lid, temperature sensor, controls, drains, and heating system for damage or leaks.
4. Clean and sanitise food-contact surfaces according to the facility's approved hygiene procedure.
5. Check water quality, steam supply, and any food-contact additives against applicable food safety requirements.
6. Verify that temperature and timing instruments are working and that guards and emergency controls are available.
7. Wear heat-resistant gloves, apron, non-slip footwear, and eye protection appropriate to the process.

How to use

1. Fill the unit with the specified water level or prepare the steam system according to the manual.
2. Start the heating system and allow the medium to reach the validated target temperature.
3. Prepare and inspect the food, removing unsuitable pieces and visible soil as required by the process.
4. Load the product basket or conveyor evenly without exceeding the rated capacity.

5. Lower or feed the product carefully to avoid splashing, burns, or mechanical trapping.
6. Start the timer when the process reaches the defined condition specified by the validated procedure.
7. Maintain the required temperature and exposure time; monitor the display and product flow throughout the cycle.
8. Remove the product using the designated basket, conveyor, or handling tool while avoiding contact with hot surfaces.
9. Transfer the blanched product promptly to the specified cooling or next processing stage.
10. Check representative product samples and record temperature, time, batch identity, and any deviations.
11. Stop heating, drain the unit where required, and allow equipment to cool safely before cleaning.

⚠ Safety

- Beware of hot water, steam, hot metal surfaces, and splashing; never reach into a running unit.
- Do not open pressurised steam equipment or remove guards while the system is operating.
- Use approved lifting tools and safe handling methods for heavy baskets or hot product loads.
- Keep electrical connections and controls protected from water and follow lockout procedures before maintenance.
- Prevent cross-contamination by separating raw, blanched, and cooled product handling areas.
- Follow the validated food safety process; blanching alone does not guarantee sterilisation or eliminate all pathogens.

Care & storage

- Clean and sanitise food-contact surfaces after production according to the approved sanitation schedule.
- Remove food residues from baskets, drains, and chamber areas without damaging sensors or seals.
- Inspect heating elements, steam lines, temperature sensors, gaskets, and controls regularly.
- Calibrate temperature and timing instruments at the prescribed intervals.
- Report leaks, inconsistent temperature, damaged guards, or control faults and arrange authorised repair.

Common mistakes to avoid

- Using an unvalidated temperature or blanching time for the product.
- Overloading the basket and causing uneven heat treatment.
- Delaying the required cooling or next processing step after blanching.
- Ignoring temperature deviations or failing to record batch process data.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.